

MD. MEHERAB HOSSEN

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📄 https://www.researchgate.net/profile/Md-Meherab-Hossen?ev=hdr_xprf

Professional Summary

Motivated Mechanical Engineering undergraduate with strong foundations in fluid mechanics, heat transfer, and computational analysis. Experienced in CAD design, CFD, and thermal simulation using ANSYS, OpenFOAM and COMSOL. Interested in energy systems, manufacturing, and applied mechanics.

Education

Ahsanullah University of Science and Technology (AUST)

April 2022 – July 2026

B.Sc. in Mechanical Engineering.

Thesis

Experimental Characterization of *Nymphaea Rubra* (Water Lily) Natural Fiber and Its Epoxy Composite through Mechanical Analysis

Ahsanullah University of Science and Technology (AUST) — July 2026

Supervisor: Dr. Fazle Rabbi — Three-Member Research Team

- Investigated water-lily natural fiber as a sustainable reinforcement material for epoxy-based polymer composites.
- Compared four fiber-extraction conditions involving 48-hour and 72-hour natural retting, with and without 6% NaOH treatment.
- Conducted ASTM-based single-fiber tensile testing using a Universal Testing Machine and analyzed load-extension and stress-strain data to determine tensile strength and Young's modulus.
- Fabricated vertically aligned water-lily-fiber/epoxy composite plates using the hand lay-up method and prepared standardized specimens through laser cutting.
- Performed tensile, flexural, Charpy impact and Brinell hardness testing, along with SEM examination of fiber-matrix bonding, voids and fiber pull-out.
- Identified 48-hour natural retting as the most effective extraction condition and obtained an average composite tensile strength of 25.5 ± 2 MPa, elastic modulus of 3.2 ± 0.29 GPa, and flexural strength of 22 ± 5.5 MPa.

Technical Skills: Composite Fabrication, Hand Lay-Up, UTM Operation, Mechanical Testing, SEM Analysis, ASTM Standards, Stress-Strain Analysis, CAD Drawing, Laser Cutting, Experimental Data Analysis

Projects

- **Gear Train for Rotary Screw Compressor** — *Designed gear system in SolidWorks and analyzed torque transmission efficiency.*
- **Belt Drive for Car Starter Motor** — *Designed and evaluated belt drive considering mechanical constraints.*
- **Water Quality Monitoring System** — *IoT-based prototype using Arduino and sensors; presented at project showcasing.*

Professional Experience

CFD Simulation Engineer — Remote

MACPROTEC Houston, Texas, USA — July 2026–Present

- Perform CFD analysis of industrial dust-collection systems, baghouses, and process pipelines using ANSYS Fluent.
- Calculate pressure drop, velocity, mass-flow rate, and pressure distribution through pipes, fittings, angular elbows, and elevator heads.
- Apply suitable turbulence and energy models to study airflow and system performance under different operating conditions.
- Analyze flow distribution inside baghouses and dust collectors to identify high-pressure-loss regions and uneven airflow.
- Develop and modify three-dimensional geometries using ANSYS DesignModeler and SolidWorks.
- Prepare meshes, define boundary conditions, run simulations, and interpret pressure, velocity, and turbulence contours.
- Support selected OpenFOAM cases and assist with the company's transition toward open-source CFD simulation.
- Present simulation results and technical observations to support design evaluation and system improvement.

Industrial Attachment

Bangladesh Industrial Technical Assistance Centre (BITAC), Dhaka

- Hands-on training in CNC machining, lathe operation, heat treatment, molding, and casting.

Skills

Simulation & Design: *SolidWorks (Advanced), ANSYS Fluent, OpenFOAM, COMSOL, LabVIEW*

Programming: *Python (Beginner), C, C++*

Tools: *MS Word, Excel, PowerPoint, LaTeX.*

Interpersonal Skills

Quick learner, Problem solving, Time management, Communication, Leadership

Awards

People's Choice Award — *Mechanical Project Showcasing (Fall 2023)*

Reference

Prof. Dr. Dewan Hasan Ahmed

Department of Mechanical & Production Engineering, AUST

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